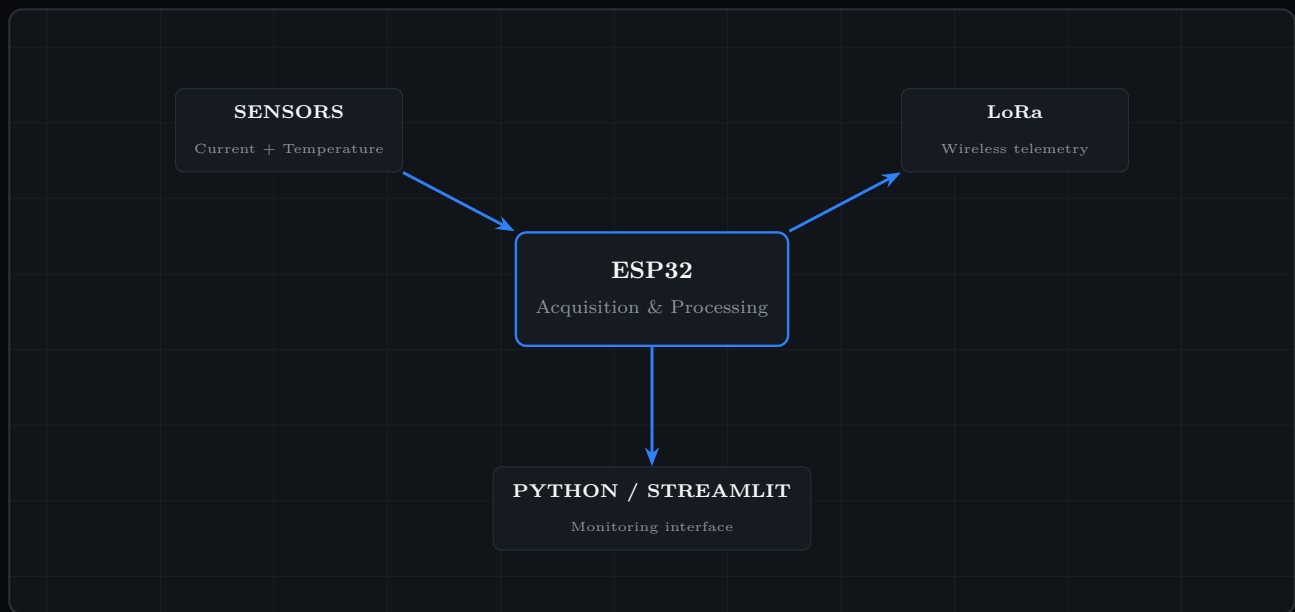


ESP32 Predictive Maintenance Node

Industrial IoT · Embedded Monitoring · LoRa · Data Visualization



ESP32

LoRa

Industrial IoT

Condition Monitoring

From physical signals to connected machine data

PROJECT SCOPE

Physical layer

Current + temperature

Embedded layer

ESP32 acquisition

Communication

LoRa telemetry

Monitoring

Python / Streamlit

The engineering problem

Industrial equipment generates measurable physical signals that can provide useful information about its operating condition. The challenge is to acquire these signals reliably, transmit them from the embedded node, and present them in a form that can be monitored and interpreted.

The proposed approach

This project implements an ESP32-based monitoring node capable of acquiring current and temperature measurements and transmitting the collected data through a LoRa communication link toward a monitoring interface developed with Python and Streamlit.

Engineering objective

The objective is to establish a connected condition-monitoring architecture that links physical measurements, embedded processing, wireless telemetry, and data visualization.

ENGINEERING CHAIN

Physical

measurements



Embedded

acquisition



Connected

telemetry

A complete monitoring path: the system starts with measurable physical quantities and ends with a software interface capable of displaying the collected information.

02

Measured parameters

01

Embedded node

01

Wireless link

A distributed monitoring architecture

System data path

The architecture separates sensing, embedded acquisition, wireless communication, and visualization. This creates a clear data path from the monitored physical system to the operator interface.

DATA FLOW

01 Physical measurements

Current and temperature signals provide the monitored data.

02 ESP32 acquisition

The embedded node acquires and handles the measurements.

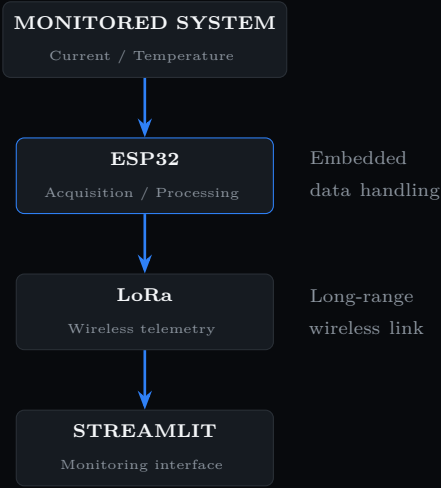
03 LoRa transmission

Collected information is sent through the wireless link.

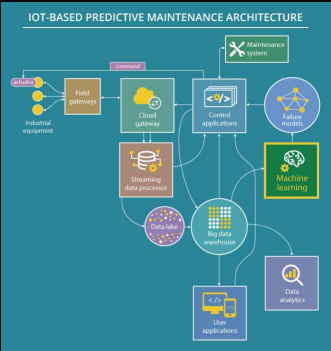
04 Monitoring interface

Python / Streamlit provides the visualization layer.

Architecture overview

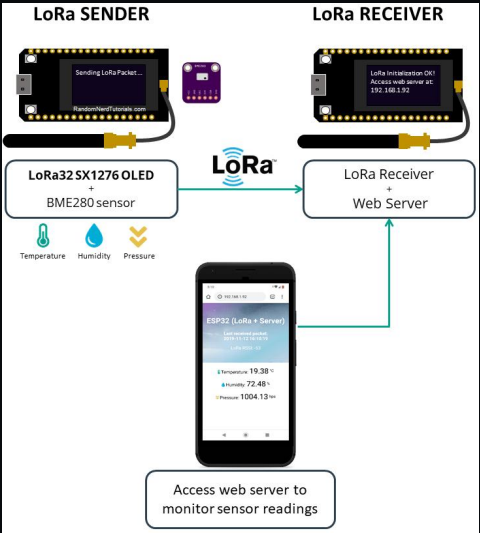


Physical implementation



Project architecture / hardware evidence.

The embedded node



ESP32 as the acquisition layer

The ESP32 forms the central embedded element of the monitoring node. Its role is to interface with the measurement layer, handle acquired data, and participate in the wireless communication chain.

Hardware implementation — project boards and embedded electronics.

Core responsibilities

- Sensor acquisition
- Local data handling
- Wireless communication
- Monitoring data transmission

ENGINEERING FOCUS

Hardware / Software Interaction

The project connects physical sensing with embedded computation and wireless data communication.

TECHNICAL STACK

ESP32

Embedded node

Sensors

Current + temperature

LoRa

Wireless telemetry

Python / Streamlit

Visualization

Monitoring the machine condition

From telemetry to visibility

The monitoring interface provides a software layer for observing the data collected by the embedded system. Python and Streamlit are used to transform the transmitted measurements into an interactive visualization environment.

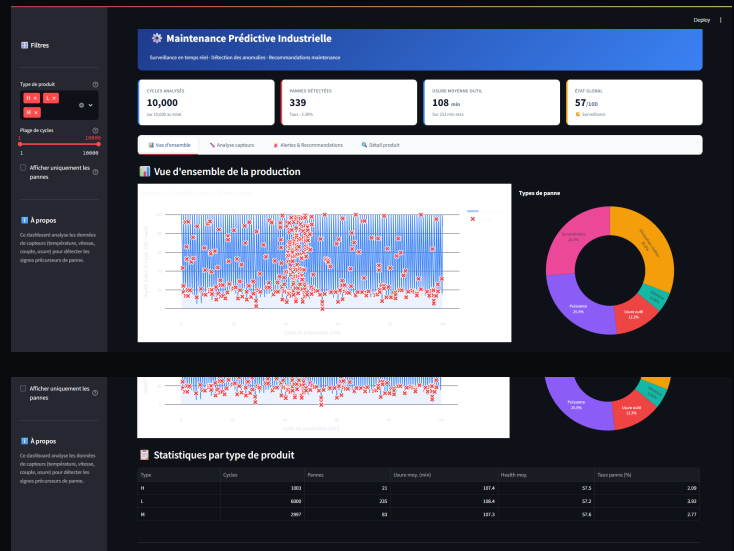
What the interface provides

- Current monitoring
- Temperature monitoring
- Graphical visualization
- Data-oriented supervision

SOFTWARE LAYER

Python + Streamlit

A lightweight interface for turning embedded telemetry into visible machine-condition information.



Monitoring dashboard — project implementation.

What this project proves

This project demonstrates the ability to connect physical measurements, embedded processing, wireless communication, and software visualization within a single monitoring architecture.

01

Embedded Systems

ESP32-based acquisition and local data handling.

02

Industrial IoT

Physical measurements connected to a remote monitoring layer.

03

Wireless Communication

LoRa-based telemetry between the embedded monitoring node and the visualization layer.

04

Data Visualization

Python / Streamlit interface for monitoring current and temperature data.

Embedded sensing · LoRa telemetry · Industrial monitoring ;